Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/08/2025
PI Name:	Bowles, Dawn Ph.D.		
Project Title:	Proteomic Signatures of Space Radiation Induced Cardiovascular Degeneration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27710-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	05/12/2016	**End Date:**	11/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2022 per NSSC information (Ed., 6/29/22). NOTE: End date changed to 5/11/2022 per L. Barnes-Moten/JSC and NSSC information (Ed., 5/10/21) NOTE: End date changed to 5/11/2021 per NSSC information (Ed., 8/12/20)		
Key Personnel Changes/Previous PI:	November 2021 report: Dr. Mark Dewhirst retired and is no longer CoInvestigator on the project.		
COI Name (Institution):	Abraham, Dennis M.D. (Duke University) Kidane, Yared Ph.D. (Wyle Laboratories, Inc.) Mao, Lan M.D. (Duke University) Moseley, Martin Ph.D. (Duke University)		
Grant/Contract No.:	NNX16AK20G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Radiation damage and the cell's attempt to repair it triggers a myriad of signal transduction pathways which alter gene, and ultimately, protein expression. Space radiation may affect biomolecules, cellular processes, and ultimately the cellular protein content (the proteome) differently than radiation present on Earth. Epidemiological analysis of terrestrial radiation exposure indicates that single high- or multiple low-dose radiation exposure can culminate in a wide array of cardiac injury and malfunction over time. Based on terrestrial data, it is believed that cardiovascular disorders may develop in astronauts from exposure to the space radiation environment. Indeed, a recent study by Yan et al. (2014), found that a single full body exposure to a low dose of proton or iron particle radiation, which somewhat mimics the space radiation environment, was sufficient to induce a significant, long term, negative effect on murine cardiovascular function. In this proposal, we take advantage of our expertise with bioinformatics analysis of cardiovascular proteomic data sets and murine cardiovascular physiology to evaluate the consequences of low dose, chronic space radiation, or mixed field space radiation on the dynamics of the cardiac proteome and to understand how the radiation induced changes relate to cardiovascular function. In doing so, we will extend Yan et al.'s work by identifying a proteomic signature that predicts the development of permanent cardiovascular degeneration from a single low dose space radiation exposure. Further, we seek to evaluate whether the proteomic signatures differ when mice experience repeated exposures of space-like radiation or mixed field space radiation. This information will lead to a mechanistic understanding of the altered cellular and molecular processes contributing to the development of cardiovascular dysfunction at the organ and organismal level in scenarios better mimicking the space radiation environment. This information is needed to predict, monitor, and prevent cardiac damage during long term space flight. Reference: Yan, X., et al., Cardiovascular risks associated with low dose ionizing particle radiation. PLoS One, 2014. 9(10): p. e110269.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Limited information is known regarding the impact of chronic low level radiation on cardiovascular molecular biology and function both terrestrially and during extended space exploration. Our research is expected to provide information in regards to terrestrial and astronaut heath. Innovative technologies that may arise from our studies may include novel biomarkers predictive of cardiovascular susceptibility to chronic low level radiation as well as countermeasures that may be employed both on Earth as well as during space exploration.
Task Progress:	The complex and inaccessible space radiation environment poses an unresolved risk to astronaut cardiovascular health during long-term space exploration missions. To model this risk, healthy male c57BL/6 mice, aged six months (corresponding to an astronaut of 34 years), were exposed to simplified Galactic Cosmic Ray (GCR5-ion; 5-ion sim) irradiation at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratories (BNL). Multi-modal cardiovascular functional assessments performed longitudinally and terminally revealed significant impairment in cardiac function in mice exposed to GCR5-ion compared to unirradiated controls, gamma irradiation, or single mono-energetic ions (56Fe or 16O). GCR5-ion-treated mice exhibited increased arterial elastance, likely mediated by disruption of elastin fibers. This study suggests that a single exposure to GCR5-ion is associated with a deterioration in cardiac structure and function that becomes apparent long after exposure, likely associated with increased morbidity and mortality. To investigate the biological basis of this dysfunction, we performed a quantitative mass spectrometry-based proteomics analysis of heart tissue (proteome and phosphoproteome) and plasma (proteome only) from these mice at eight months post-radiation. Differentially expressed proteins (DEPs) for irradiated versus sham irradiated samples (fold-change $\geq$ 1.2 and an adjusted p-value of $\leq$ 0.05) were identified for each proteomics data set. For the heart proteome, there were 87 significant DEPs (11 up-regulated and 76 down-regulated); for the heart phosphoproteome, there were 443 significant differentially phosphorylated peptides (123 up-regulated and 320 downregulated); and, for the plasma proteome, there were 188 significant DEPs (24 up-regulated and 164 down-regulated). A Gene Set Enrichment Analysis (GSEA) technique that assesses canonical pathways from BIOCARTA, KEGG, PID, REACTOME, and WikiPathways revealed significant perturbation in pathways in each data set. For the heart proteome, 2812 pathways were altered (1069 upregulated and 1743 down-regulated); for the plasma proteome, there were 1576 pathways altered (874 upregulated and 702 downregulated); and, for the phosphoproteome, there were 742 pathways affected. Pathways related to inflammation were the most highly perturbed in the heart and plasma. In line with sustained inflammation, neutrophil extracellular traps (NETS) were demonstrated to be increased in GCR5-ion irradiated hearts at 12-month post irradiation. NETS play a fundamental role in combating bacterial pathogens, modulating inflammatory responses, inflicting damage on healthy tissues, and escalating vascular thrombosis. These findings suggest that a single exposure to GCR5-ion results in long-lasting changes in the proteome and that these proteomic changes can potentiate acute and chronic health issues for astronauts, such as what we have described with late cardiac dysfunction in these mice.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Kidane YH, Lee FH, Smith MF, Wang C, Mirza JB, Sharma S, Lobo AA, Dewan KC, Chen J, Diaz TE, Pla MM, Foster MW, Bowles DE. "Proteomic and phosphoproteomic characterization of cardiovascular tissues after long term exposure to simulated space radiation." Front Physiol. 2024 Apr 16;15:1248276. https://doi.org/10.3389/fphys.2024.1248276 ; PMID: 38699144; PMCID: PMC11063234 , Apr-2024
Articles in Peer-reviewed Journals	Bishawi M, Lee FH, Abraham DM, Glass C, Blocker SJ, Cox DJ, Brown ZD, Rockman HA, Mao L, Slaba TC, Dewhirst MW, Truskey GA, Bowles DE. "Late onset cardiovascular dysfunction in adult mice resulting from galactic cosmic ray exposure." iScience. 2022 Apr 15;25(4):104086. https://doi.org/10.1016/j.isci.2022.104086 ; PMID: 35378858; PMCID: PMC8976132 , Apr-2022
Articles in Peer-reviewed Journals	Pla MM, Lee FH, Smith MF, Lobo AA, Diaz TE, Cockrell AS, Bowles DE. "Double-stranded RNA and reverse transcriptase signatures of endogenous retroviruses after exposure to spaceflight environment related stressors." Life Sci Space Res. 2026 Jun 19. Online ahead of print. https://doi.org/10.1016/j.lssr.2026.06.005 , Jun-2026